Kevin Le

Computer Engineer

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Education

University of California, Davis — Davis, CA
Master of Science in Computer Engineering | GPA: 4.0 — *Sep 2024 – Dec 2026*

University of California, Davis — Davis, CA
Bachelor of Science in Computer Engineering | GPA: 3.75 — *Sep 2021 – Jun 2025*

- Deans Honor's List — Tau Beta Pi Scholar

Relevant Skills

Languages: C/C++, Python, Rust, Verilog, RISC-V, Javascript
Developer Tools: Git, Kubernetes, Docker, AWS, Nsight Compute (NCU)
Miscellaneous: CUDA, ROCm, PyTorch, Triton, Linux, OpenMPI, OpenMP, OpenCL, FPGA

Work Experience

AggieWorks: RoomU Team — *Sep 2023 – Dec 2024*
Software Engineering Intern

- Updated a Typescript code base using a React framework to implement critical messaging features for a social network app aimed to connect students looking for roommates
- Worked within a small cross-functional team of developers and designers to reform the application's features and improve user retention
- Interfaced with an SQL database to support page-caching protocols that reduced page access times nearly 65% in order to improve user experience

UC Davis College of Engineering — *Sep 2024 – Mar 2025*
Lab Teaching Assistant

- Responsible for leading lab sessions for two separate morning section labs for the Circuits course in the Electrical Engineering department
- Taught proper oscilloscope operation, breadboard best practices, and conducted lab demos to explain lecture materials

Projects

Neural Network in CUDA

- Prototyped and trained a neural network architecture using PyTorch to classify a large dataset of handwritten digits
- Reimplemented and profiled the same architecture on CUDA kernels in order to discern inefficiencies in SM utilization and memory-transfer throughput
- Optimized from naive kernel implementations in order to achieve comparable training time with the Pytorch model as well as equivalent model performance with a 98% classification accuracy

2D Map Reconstruction

- Synthesized a computer vision solution using a C++ localization algorithm for accurate map generation of a building's floor plan based on camera data
- Trained an object detection model in PyTorch (YOLO), which could identify and label particular landmarks such as doors or tables that the camera comes across during its survey of the floor
- Implemented a Python script that synchronized trajectory data with the detection model's output to create a 2D floor layout with marked landmarks

Distributed Computing using MPI

- Created a C program that calculates and visualizes a selected region of the famous Mandelbrot Set using multiple compute nodes on the school server
- Optimized the serial implementation via the use of the MPI via the OpenMPI communication library to connect different nodes on the server to achieve a 300% decrease in compute time from 15 seconds to just 5 seconds
- Wrote a Shell Script that automated the employment of available computers in UC Davis's server room to increase the node count and computing power of the program